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A PALAEOMAGNETIC STUDY OF SOME BASALTOIDS AND ORES FROM THE PONTIC RANGES, NORTHERN TURKEY: PALAEOGEOGRAPHIC IMPLICATIONS

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ABSTRACT

Güner, M., 1982. A palaeomagnetic study of some basaltoids and ores from the Pontic Ranges, northern Turkey: palaeogeographic implications. *Tectonophysics*, 90: 309–333.

The palaeomagnetism of basic rocks and sulphide ores has been studied in the Küre area, Pontic Ranges, Turkey. Progressive alternating-field demagnetization revealed a characteristic remanent magnetization in all investigated rock types except a dacite. The following virtual geomagnetic poles were obtained:

Basalt and quartz diabase (oldest): $D = 59°$, $I = +66°$, $\alpha_{95} = 4.8$, pole 49°N, 93°E.

Diabase: $D = 210°$, $I = -15°$, $\alpha_{95} = 15.0$, pole 47°N, 167°E.

Massive sulphide ores: $D = 107°$, $I = +63°$, $\alpha_{95} = 8.7$, pole 18°N, 80°E.

Peridotite: $D = 131°$, $I = +54°$, $\alpha_{95} = 10.9$, pole 2°S, 72°E.

Amphibolitized diabase (youngest): $D = 293°$, $I = +59°$, $\alpha_{95} = 12.6$, pole 40°S, 145°E.

The longidutinal difference in pole positions between the oldest and the youngest rocks is interpreted as being due to a post-Permian counterclockwise rotation of the studied region in relation to the African continent. In addition, there are indications of local rotational movements within the Küre area.

INTRODUCTION

This paper presents a palaeomagnetic investigation of rocks and ores from the Küre area, Pontic Ranges, northern Turkey. The studied units comprise basalts (massive flows and pillows), massive sulphide ores, and various dykes consisting of quartz diabase, diabase, peridotite, and amphibolitized diabase. Altogether 113 specimens from 42 localities have been measured. The palaeomagnetic work was carried out at the Palaeomagnetic Laboratory of the Geological Institute, Lund.

The object of the present investigation is to compare Permo-Jurassic results from the Küre area with those from other parts of Turkey, Armenia, Minor Caucasus, Africa, the Arabian Shield, and Europe. It is also of interest to see whether the palaeomagnetic data for the various units can shed light on their relative movements

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within the faulted Alpine orogenic belt of this area, and to study the possibility of arranging the events in chronological order.

GEOLOGY

Küre is situated at 41.81°N, 33.72°E, 30 km from the Black Sea coast, on the western flank of the Pontic Ranges (Fig. 1), which extend along the coast from Zonguldak to Rize. The investigated area lies about 70 km to the north of the great zone of dextral strike-slip faults, the North Anatolian fault zone, which goes from west to east straight across northern Turkey from the Dardanelles to Van Gölü (Fig. 1). Moreover, according to Pavoni (1961), one of the many tectonic lineaments belonging to this large transcurrent fault forks in Ladik–Kavak and passes east of Küre to Inebolu and into the Black Sea (Fig. 1).

The geology of the area and the massive sulphide deposits at Küre have been described elsewhere (Güner, in press, a, b). A large succession of submarine basalts, generally massive flows at the base, that is upwards gradually followed by pillows, pillow breccias, and tuffaceous chloritic masses, has intruded into the eugeosynclinal accumulations of subgraywacke and black shale. The latter is estimated to be Permian (Güner, in press, a). The basalt complex, in turn, has been intruded by numerous dykes of diabases, dacite, and peridotite (Fig. 2). The massive sulphide ores occur as hydrothermally deposited emplacements along weak zones in the basalt succession. The dykes and the sulphide ores are well distributed both in time and in

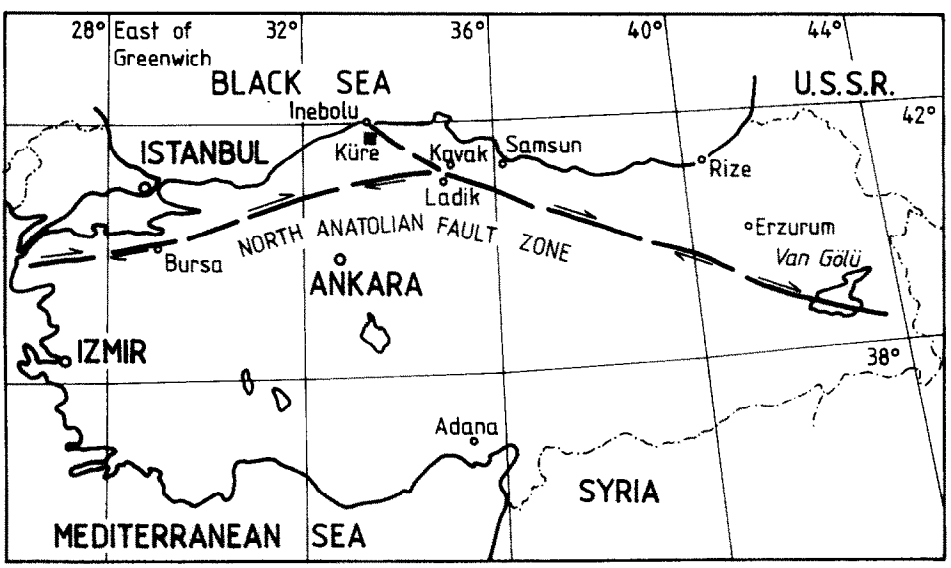


Fig. 1. Sketch-map of Turkey, showing the North Anatolian fault zone.

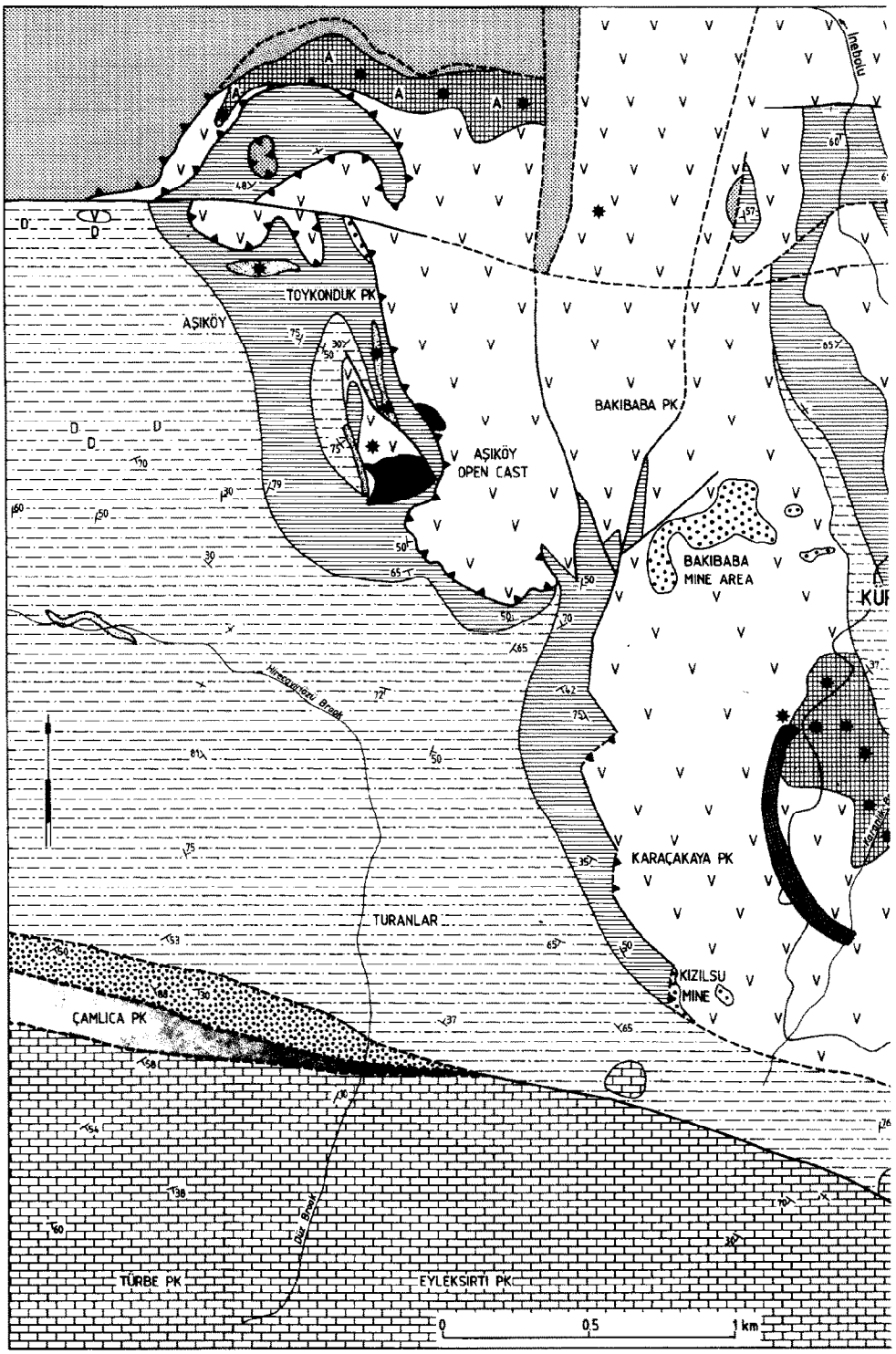
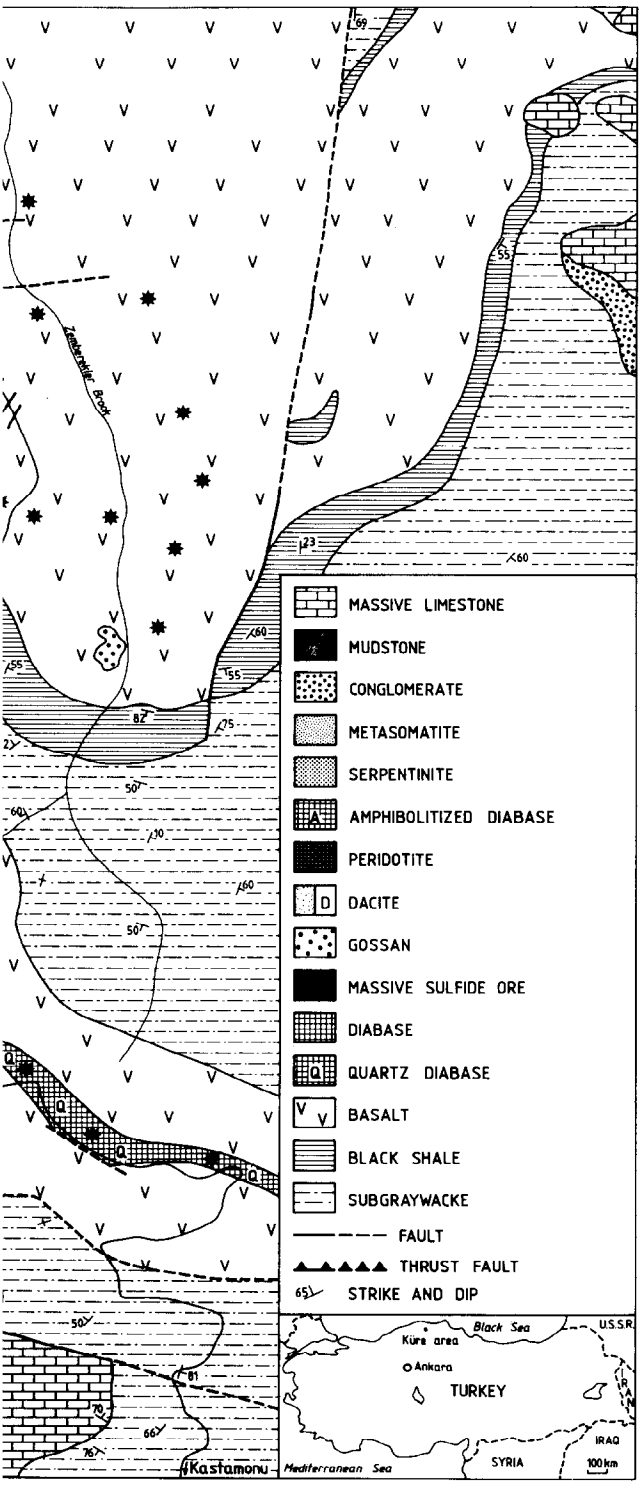


Fig. 2. Petrological map of the Küre area. * = sampling site.


MASSIVE LIMESTONE
MUDSTONE
CONGLOMERATE
METASOMATITE
SERPENTINITE
AMPHIBOLITIZED DIABASE
PERIDOTITE
DACITE
GOSSAN
MASSIVE SULFIDE ORE
DIABASE
QUARTZ DIABASE
BASALT
BLACK SHALE
SUBGRAYWACKE
FAULT
THRUST FAULT
STRIKE AND DIP
Kastamonu
Black Sea
U.S.S.R.
Kure area
Ankara
TURKEY
IRAQ
SYRIA
Mediterranean Sea
100km

type of rock and give information on the tectonic development of this mobile belt. Presumably, the Alpine movements of Early Kimmerian, Triassic age, had generated a series of N–S trending faults, which then acted as main channel ways for the ore solutions. Ketin (1962) stated that the orogenic movements in northern Anatolia during Permo-Jurassic phases had been violent. It is most probable that, at the end of the Jurassic, the Küre area was raised above sea level as an island, whereas its surroundings were still under water, which is indicated by the Cretaceous formation. According to Brinkmann (1976), the Pontic Ranges were formed in two phases. The older mountains are represented by the inner zones of the range and are largely incorporated into the outer Pontides. The youngest formation in the investigated area, a massive limestone, has been dated by finds of Late Jurassic fossils (Kovenko, 1944; Sarican, 1968).

According to present knowledge (Güner, in press, a, b) the stratigraphy of the area is as follows:

Youngest:	Uplift of the Küre area above sea level
	Folding of eugeosynclinal accumulations
	Faulting of non-eugeosynclinal sediments
Late Jurassic:	*Formation of massive limestone*
	Sedimentation of mudstone
	Formation of the conglomerates
	Thrust faulting of the different units
	Development of metasomatic rock
	Serpentinization
	Intrusion of amphibolitized diabase
	Intrusion of peridotite
	Intrusion of dacite
	Formation of the sulphide ores from hydrothermal solutions
	Development of major faults and thrust faulting
	Intrusion of diabase
	Intrusion of quartz diabase
	Extrusion of the rocks of the basalt complex
	Isoclinal folding of the eugeosynclinal accumulations
Permian (?):	*Sedimentation of black shale*
Oldest:	*Sedimentation of subgraywacke*

SAMPLING AND MEASUREMENT

Only handsamples were collected. Care was taken to sample only in areas not affected by the younger tectonic movements (Fig. 2). 113 specimens were drilled from 46 oriented samples and sawn into cylinders 25 mm in diameter and 25 mm in length. The directions and intensity of remanent magnetization of each specimen

TABLE I

Palaeomagnetic data of measured rock types from Küre area *

Rock type	n	N	NRM			S	After AF demagnetization					Pole position	
			D	I	α_{95}		D	I	R	k	α_{95}	lat.	long.
Basalt	14	34	33.4°	+72.4°	10.2	32–5562	64.0°	+65.4°	32.2	18.6	5.8	45.4°N	93.5°E
Quartz diabase	3	9	21.4°	+75.8°	18.1	1153–3649	51.2°	+69.3°	8.6	19.5	11.9	54.4°N	87.4°E
Diabase dyke	6	15	216.9°	+33.7°	47.4	27–2110	210.2°	−15.2°	13.1	7.3	15.0	46.7°N	166.9°E
Massive sulphide ore	7	20	53.8°	+70.0°	9.6	11– 70	107.0°	+63.0°	18.7	15.0	8.7	18.1°N	79.5°E
Dacite dyke	4	12	309.3°	+41.0°	53.7	4– 12	266.4°	+5.1°	9.6	4.4	23.1	0.9°N	127.9°E
Peridotite dyke	4	13	76.3°	+64.3°	21.8	228–2868	130.9°	+53.6°	12.2	15.2	10.9	1.6°S	72.3°E
Amphibolitized diabase	4	10	290.3°	+70.8°	19.6	44– 296	292.8°	+58.7°	9.4	15.4	12.6	40.3°S	144.8°E

* N.R.M. = Natural Remanent Magnetization; n = number of sample localities; N = number of specimens measured; D = declination of remanent magnetization; I = inclination of remanent magnetization; α_{95} = semi angle of cone of 95% confidence for mean direction; S = susceptibility (μG/Oe); R = length of the resultant of the N unit vectors; $k = (N-1)/(N-R)$ = estimate of Fisher's (1953) precision parameter K.

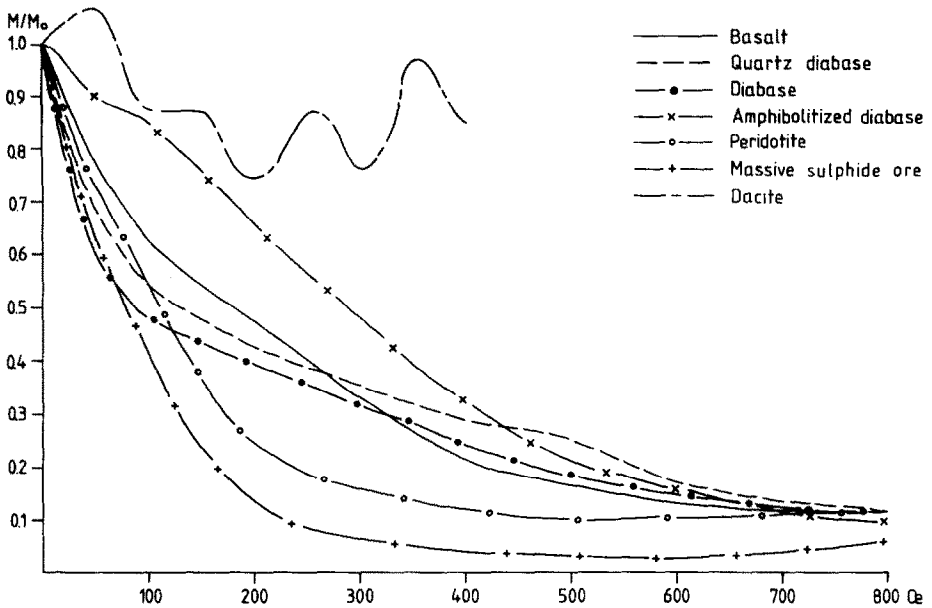


Fig. 3. Plot of M/M_0 versus the demagnetizing field. M_0 = intensity of the initial remanence, M = intensity measured after demagnetization in field.

was measured with a Digico Balanced Fluxgate Rock Magnetometer.

Systematic alternating-field demagnetization was carried out in an attempt to isolate the hardest component of magnetization. For each specimen, the demagnetizations were done stepwise measurements after each step. In Fig. 3, the variations of the ration M/M_0 versus the demagnetizing field demonstrate that all the curves, with the exception of that for the dacite, have uniform relative decay patterns, which indicates the presence of minerals with low coercivity. MDF is about 200 Oe. It is seen from the intensity curves that all the specimens of the various units retain ca. 15% of their initial moment up to 800 Oe peak fields. During demagnetization, a stable consistent direction of the magnetic vector was generally obtained in fields as low as 50 Oe. The data obtained by the measurements are listed in Table I. The magnetic directions before and after demagnetization are plotted in stereograms (Figs. 4–6).

Basalt

The 34 specimens of massive flows and pillows were collected from 14 localities. The intensity of the NRM varies from about 2 μG to 2 mG. Some of the measured specimens displayed a remaining stable remanence after demagnetization in alternating fields between 50 and 100 Oe, indicating that a small secondary component has

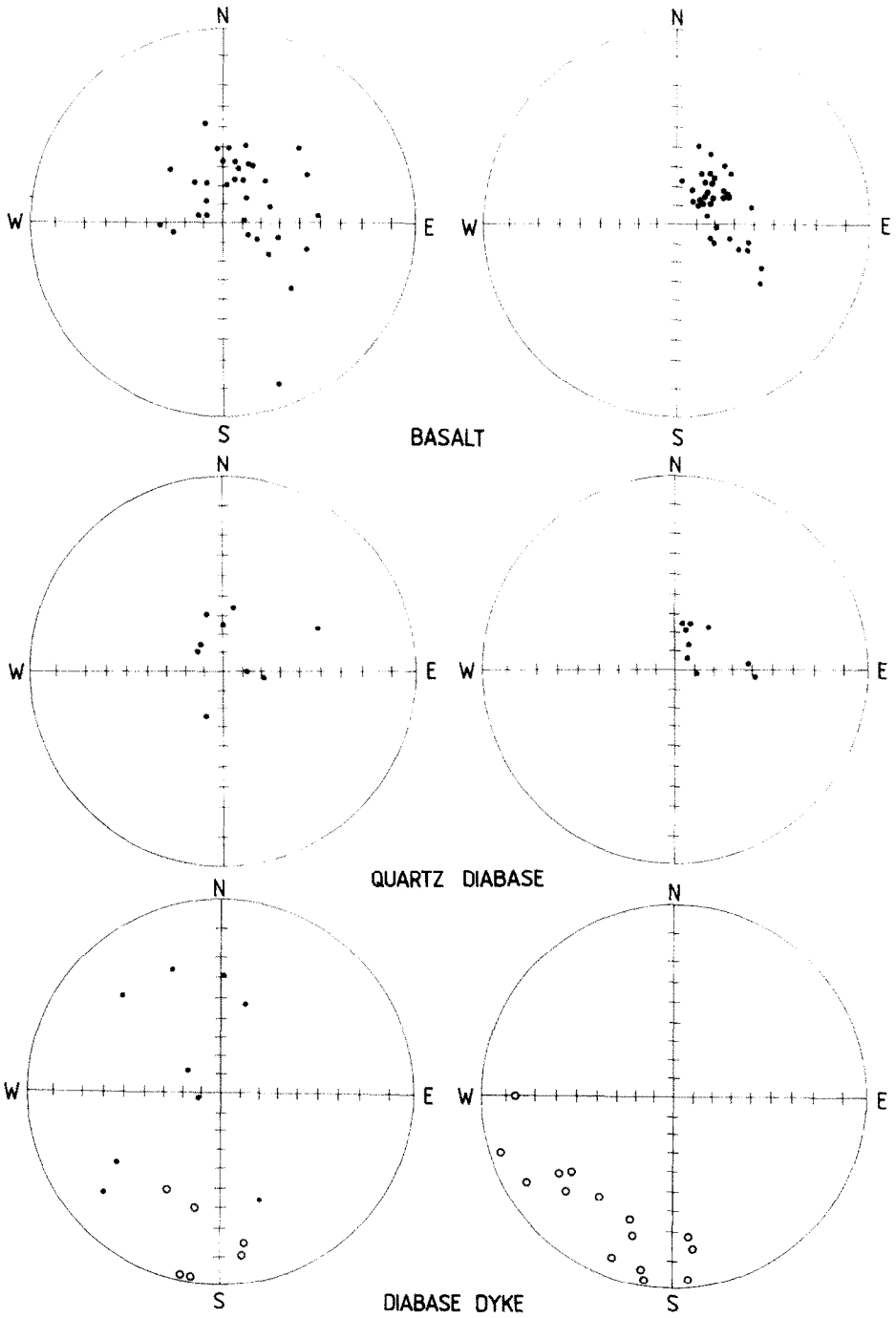


Fig. 4. Directions of remanent magnetization before (left) and after demagnetization to a stable end-point (right).

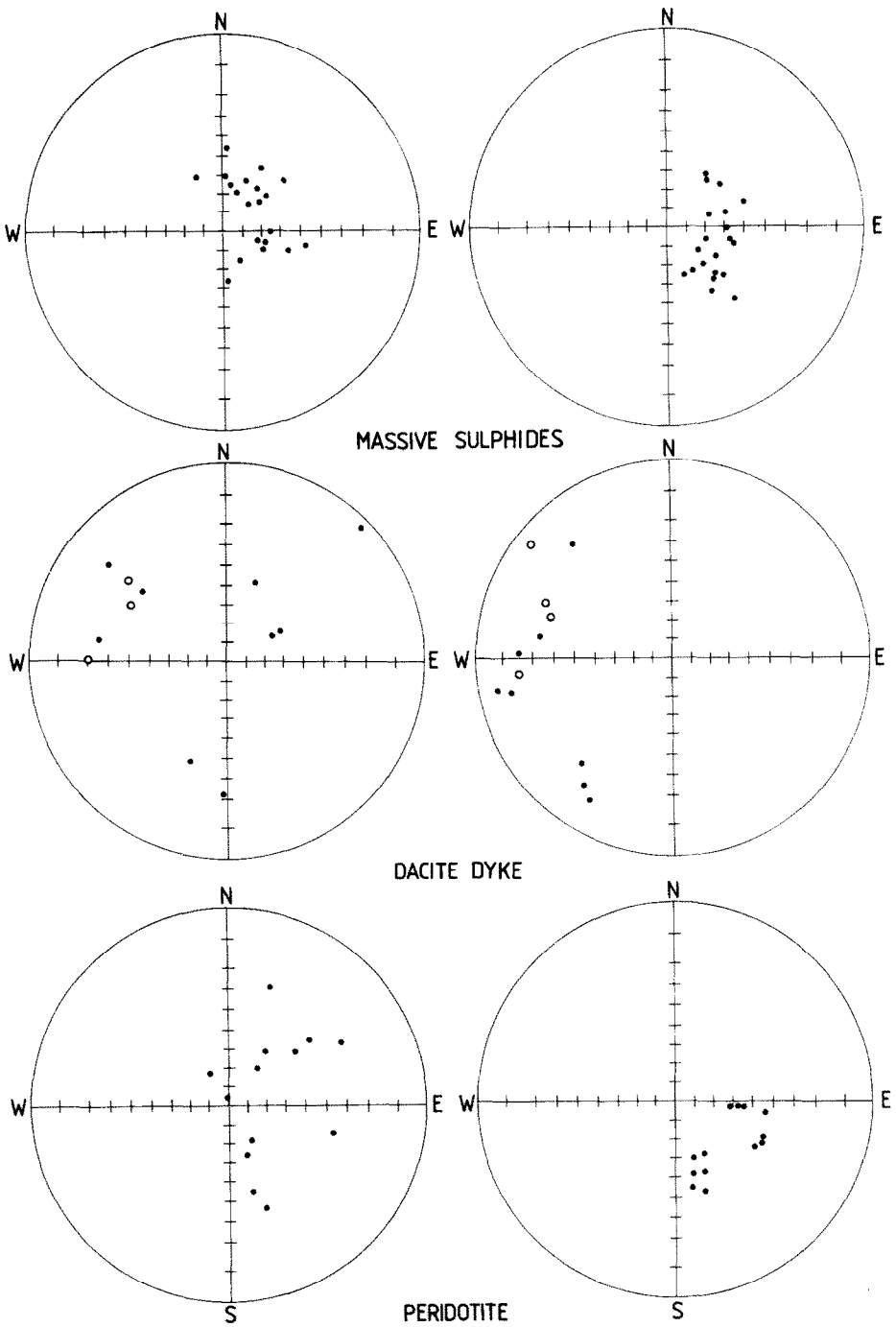


Fig. 5. Directions of remanent magnetization before (left) and after demagnetization to a stable end-point (right).

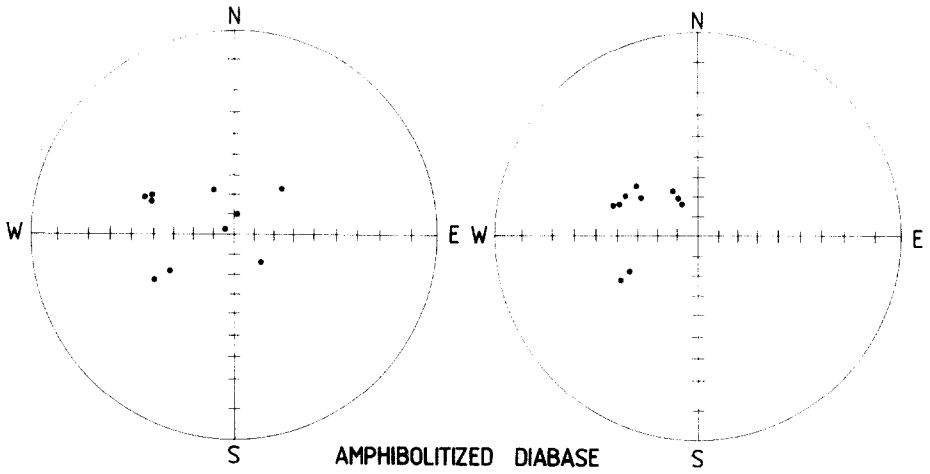


Fig. 6. Directions of remanent magnetization before (left) and after demagnetization to a stable end-point (right).

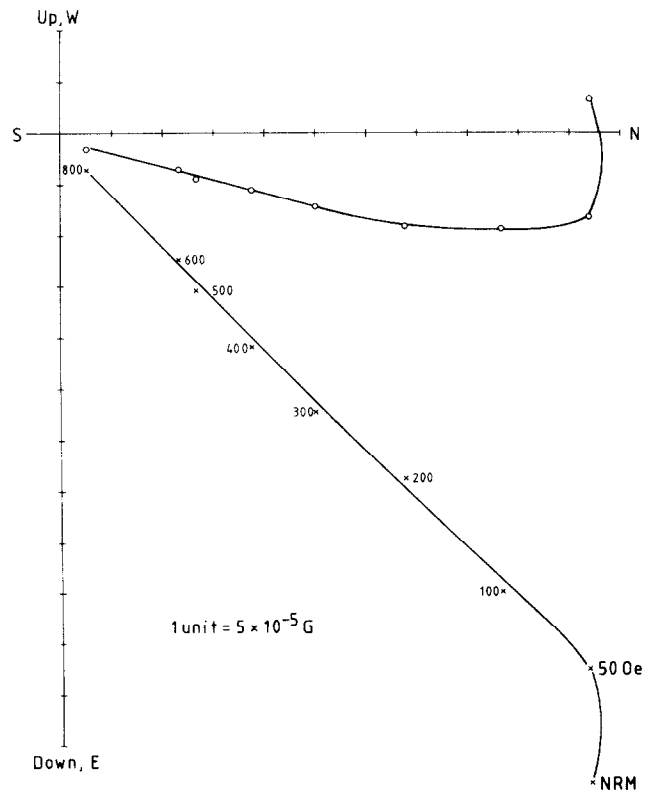


Fig. 7. Vector end-point diagram of NRM demagnetization of a typical basalt sample.

been eliminated during the first step (Fig. 7). In alternating fields above 400 Oe, they generally became unstable. Those of high initial intensity displayed a rapid initial decay in intensity during demagnetization, which suggests that the minerals that carry the NRM have a low coercivity. A sample from a locality within a fault system adjacent to the sulphide ore was strongly chloritized, that means affected by hydrothermal ore solutions. Specimens from that sample displayed different magnetic directions during successive demagnetization treatment. This is probably due to the presence of large amounts of fine secondary magnetite grains (Fig. 8A). Also samples from three other localities, collected at intrusive rock contacts, contained a stable remanence up to 500 Oe peak field, but presented a large scatter in directions between the individual samples. Sample B in Fig. 8, collected within the Bakibaba mine area at the contact with a peridotite dyke, was presumably first affected by ore

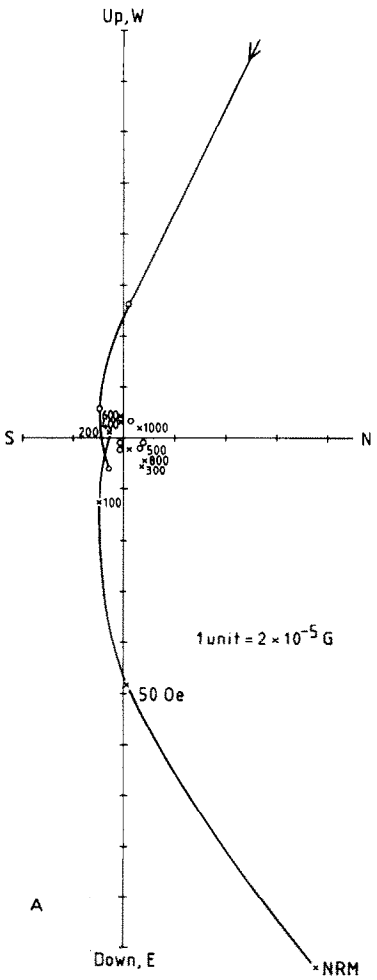


Fig. 8A. For legend see p. 322.

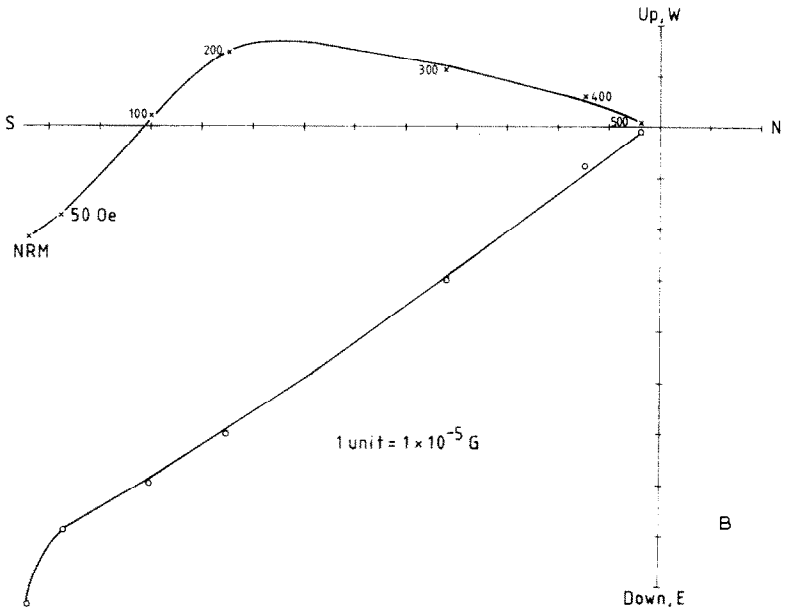


Fig. 8. Vector end-point diagram of NRM demagnetization of two basalt samples displaying different magnetic directions. Sample A is affected by ore solutions, B is from contact with a peridotite dyke.

solutions and later by the peridotite intrusion. The opaque minerals in the basalt are predominantly titaniferous magnetite and ilmenite represented by a reticulated network of leucoxene lamellae. Other opaques are chrome spinel, pyrite with martite rims, chalcopyrite, and haematite. The magnetic directions are shown in Fig. 4.

Quartz diabase

One dyke of quartz diabase was sampled. It has the same magnetic directions and demagnetizing characteristics as the basalt. The specimens have initial intensities varying between 0.5 and 2 mG and with 50% left after demagnetization at 200 Oe peak field (Fig. 3). No marked secondary components have been observed and the measured specimens seem to retain a stable remanence during demagnetization up to 800 Oe (Fig. 9). The opaques consist of magnetite grains forming skeletally reticulated crystals with a network of altered lamellae of leucoxene and occurring also as intergrowths with ilmenite plates. Some martite grains form pseudomorphs after pyrite.

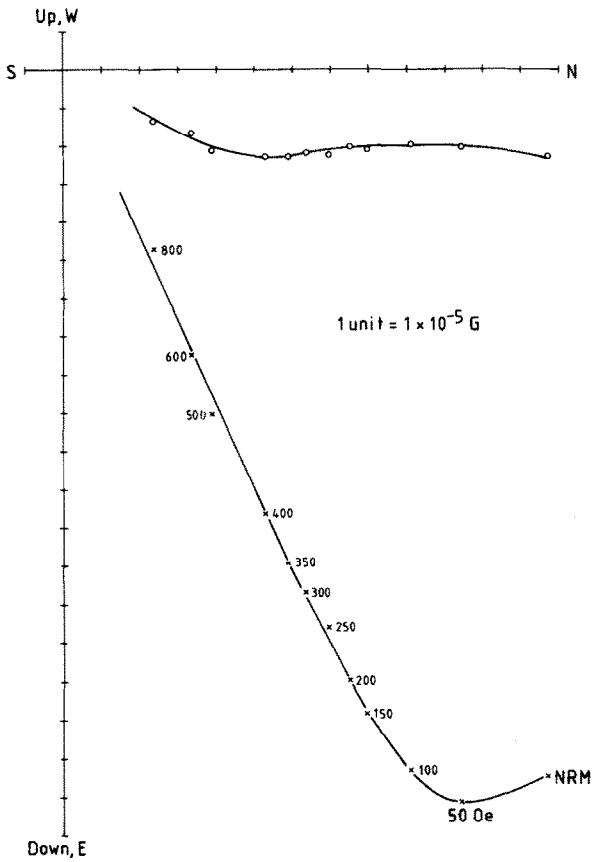


Fig. 9. Vector end-point diagram of NRM demagnetization of a typical quartz diabase sample.

Diabase

Most of the diabase specimens revealed a systematic change in direction during demagnetization. Specimens which originally gave directions with positive inclinations from 20° to 80° moved systematically to stable end-points with negative inclinations of about 10° (Fig. 4). The specimens with positive inclinations of initial NRM had commonly high intensities, which in most cases decayed rapidly in fields between 50 and 100 Oe. This suggests that the NRM consisted of two components. One very soft component due to the present geomagnetic field was eliminated early in the alternating-field treatment between 100 and 200 Oe, while another stable component was retained. In other cases, specimens with negative initial inclinations retained a larger fraction of their initial intensities of NRM to about 500 Oe. The initial NRM intensities of the specimens varied between 6 μG and 500 μG. The

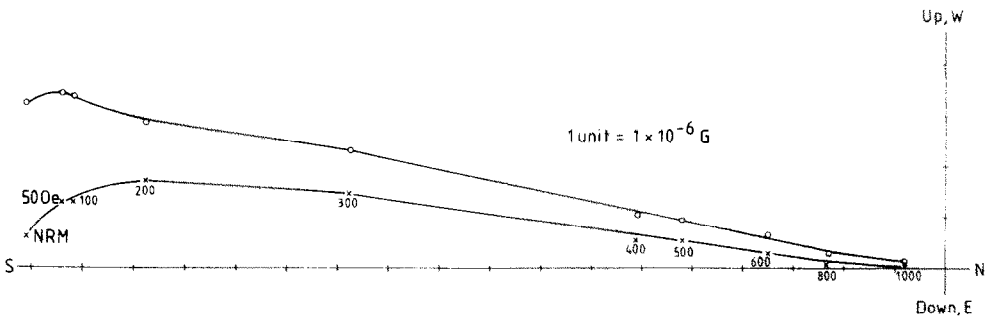


Fig. 10. Vector end-point diagram of NRM demagnetization of a typical diabase dyke sample.

opaque minerals are magnetite, pyrite, chalcopyrite, ilmenite, and haematite. The iron oxides occur as discrete grains in the matrix and in diallage clinopyroxene but are also associated with uralitization. The ilmenite contains some minute inclusions of haematite.

Massive sulphide ore

The specimens are from three ore bodies in the Bakibaba and Asiköy mines. They had rather low intensities of initial NRM, that varied from 5 to 15 μG. The sulphides show insignificant changes in the mean direction during demagnetization up to 2200 Oe (Fig. 5). The measured specimens contain 95–100% sulphide minerals, the remainder being a gangue of quartz. These sulphides consisted of pyrite, chalcopyrite, bornite, covellite, sphalerite, digenite, marcasite, tennantite, carrollite, idaite, and galena. Iron oxides or pyrrhotite could not be detected in any of more than a hundred studied polished sections. Thus, the carriers of NRM are not positively identified, but the demagnetization curves suggest fine-grained magnetite.

Dacite

The twelve specimens from the three dykes in Asiköy had very low NRM values, from 30 to 300 nG, which is close to the noise level of the instrument used. As is evident from Fig. 3, there is no systematic correlation between the M/M_0 ratio and increased demagnetization level. Also, the results from the various specimens have widely scattered directions (Fig. 5), with large α_{95} and low k values (Table I). On account of these circumstances they are considered less reliable and put between brackets in Table II. In the polished sections, no opaque minerals have been observed. However, the surface exposures are strongly oxidized and contain masses of limonite.

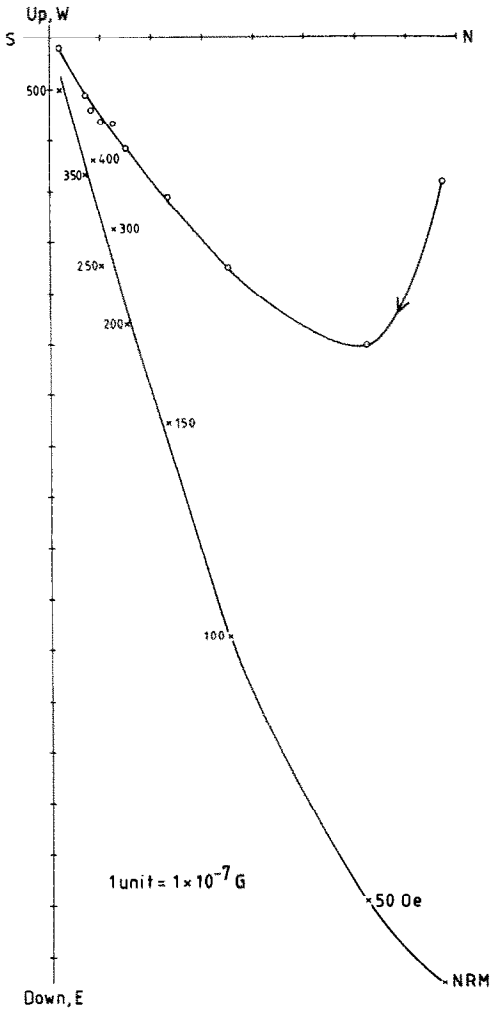


Fig. 11. Vector end-point diagram of NRM demagnetization of a typical massive sulphide sample.

Peridotite

Some of the peridotite specimens had an insignificant secondary component that was easily eliminated after alternating-field treatment in a 50 Oe peak field. Besides, there was a remaining stable component (Fig. 12). In successively higher demagnetization fields, from about 500 Oe upwards, the directions became scattered (Fig. 12). However, in spite of the considerable degree of serpentinization in the rock, there is no marked spreading in mean magnetic directions between the initial NRM and the stable end-points below a 400 Oe peak field (Fig. 5). The intensities of the initial

TABLE II

Some ancient pole positions derived from palaeomagnetic investigation on Permo-Jurassic rocks in Europe, Africa, Armenia, Minor Caucasus, the Arabian Shield and Turkey

Nr.	Continental unit and locality	Age	N	α_{95}	Pole position	References
1	Stable Europe. Oslo region, Norway	Lower Permian	484	2.8	47°N 157°E	Van Everdingen (1960)
2	Stable Europe (mean)	Upper Permian	–	–	50°N 151°E	Van der Voo and French (1974)
3	Europe. Barrot redbeds, France	Upper Permian	35	2.5	47°N 147°E	Van den Ende (1970)
4	Stable Europe (mean)	Lower Triassic	–	–	56°N 135°E	Van der Voo and French (1974)
5	Stable Europe	Triassic	–	6.0	50°N 150°E	McElhinny (1973)
6	Stable Europe	Triassic	–	10.0	51°N 144°E	Phillips and Forsyth (1972)
7	Stable Europe (mean)	Upper Triassic	–	–	68°N 137°E	Van der Voo and French (1974)
8	Europe. Northern Pyrenees, France	Late Triassic	7	7.5	62°N 114°E	Girdler (1968)
9	Stable Europe (mean)	Lower Triassic	–	–	79°N 161°E	Van der Voo and French (1974)
10	Europe. Northern Pyrenees, France	Lower Jurassic	8	6.5	65°N 143°E	Girdler (1968)
11	Armenia. Combined porphyrites	Jurassic	–	–	53°N 115°E	Irving et al. (1976)
12	Armenia. Combined vol./sed. rocks	Midd. Jurassic	262	–	71°N 151°E	Sirunyan (1975)
13	Minor Caucasus. Combined vol./sed. rocks	Midd. Jurassic	430	–	70°N 177°E	Khramov (1973)
14	Turkey. Redbeds, Amasra	Permian	15	18.0	11°S 102°E	Gregor and Zijderveld (1964)
15	Arabian Shield. Lebanon	Upper Jurassic	20	3.0	1°N 120°E	Van Dongen et al. (1967)

16	Arabian Shield. Lebanon	Upper Jurassic	–	10.5	5°S	119°E	Gregor and Nairn (1971)
17	Africa. Ecca sandstone, Tanganyika	Permian	11	7.1	58°S	74°E	Nairn (1964)
18	Africa. Ecca sandstone, Tanganyika	Permian	9	16.0	38°S	70°E	Opdyke (1964)
19	Africa. Taru Grit, Kenya	Permian	8	16.4	0°	87°E	Nairn (1959)
20	Africa. Maji Ya Chumvi, Kenya	Permian	5	11.0	4°N	150°E	Nairn (1959)
21	Africa. Redbeds/trachyandesite, Morocco	Permian	87	12.1	32°S	62°E	Martin et al. (1978)
22	Africa (mean)	Upper Permian	–	–	36°S	55°E	Van der Voo and French (1974)
23	Africa. Nuanetsi lavas, Rhodesia	Triassic	6	–	29°S	68°E	McElhinny et al. (1968)
24	Africa (mean)	Lower Triassic	–	–	48°S	57°E	Van der Voo and French (1974)
25	Africa. Argana redbeds, Morocco	Upper Triassic	22	12.0	51°S	71°E	Martin et al. (1978)
26	Africa. Karroo lavas, Rhodesia	U. Tri.-L. Jur.	–	8.0	57°S	84°E	McElhinny et al. (1968)
27	Africa (mean)	Lower Jurassic	–	–	62°S	83°E	Van der Voo and French (1974)
28	Africa. Imiter basalt, Morocco	Jurassic	6	23.8	53°N	149°E	Martin et al. (1978)
29	Africa. Imiter massif, Morocco	Midd. Jurassic	6	20.6	4°S	99°E	Martin et al. (1978)
30	Turkey. Küre basalt and quartz diabase		43	4.8	49°N	93°E	This paper
31	Turkey. Küre diabase dyke		15	15.0	47°N	167°E	This paper
32	Turkey. Küre massive sulphide ores		20	8.7	18°N	80°E	This paper
33	Turkey. Küre dacite dyke *		12	23.1	(1°N	128°E)	This paper
34	Turkey. Küre peridotite dyke		13	10.9	2°S	72°E	This paper
35	Turkey. Küre amphibolitized diabase		10	12.6	40°S	145°E	This paper

* Entered in brackets because of very low intensity and corresponding large scatter in direction.

NRM were 140 μG to 2 mG. The opaques are dominated by ilmenite and chrome spinel with additional magnetite, pyrite, pyrrhotite, and chalcopyrite. Most of the magnetite occurs in the form of fine grains in crack fillings and along the borders of the former olivine crystals.

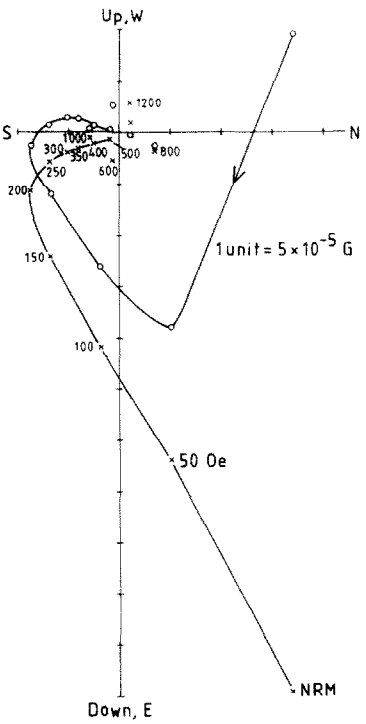


Fig. 12. Vector end-point diagram of NMR demagnetization of a typical peridotite sample.

Amphibolitized diabase

A few specimens exhibited the presence of a soft viscous magnetic component which was eliminated by treatment in a 50 Oe alternating magnetic field (Fig. 13). Most of the specimens demonstrated small changes in magnetic directions during demagnetization and thus the initial NRM was identical with the stable remanence (Fig. 6). In spite of the "amphibolitization", all specimens had very stable remanence in successively higher demagnetization fields up to 1500 Oe. In the contact with metasomatite at the western end of the dyke, the rock is medium-grained, medium bluish-gray. Its basaltic hornblende seems to be of deuteric or late magmatic origin, whilst in the rest of the dyke the clinopyroxene has been almost completely uralitized. No differences in magnetic directions were found between the western

contact rock and the main part of the dyke. This indicates that the amphibilitization had been contemporaneous with the intrusion of the dyke. The slow decline of the intensity curve (Fig. 3) indicates the presence of minerals with a somewhat stronger coercive force, that retain ca. 40% of their initial moment up to 400 Oe. This magnetic property is largely due to the presence of haematite in the rock. The intensities of initial NRM varied between 30 and 90 μG. Ilmenite, which occurs as the predominant opaque, has almost completely changed to leucoxene, which often encloses minute relicts of ilmenite and haematite in intergrowth positions. The pyrite grains are rimmed by martite with minute inclusions of chalcopyrite.

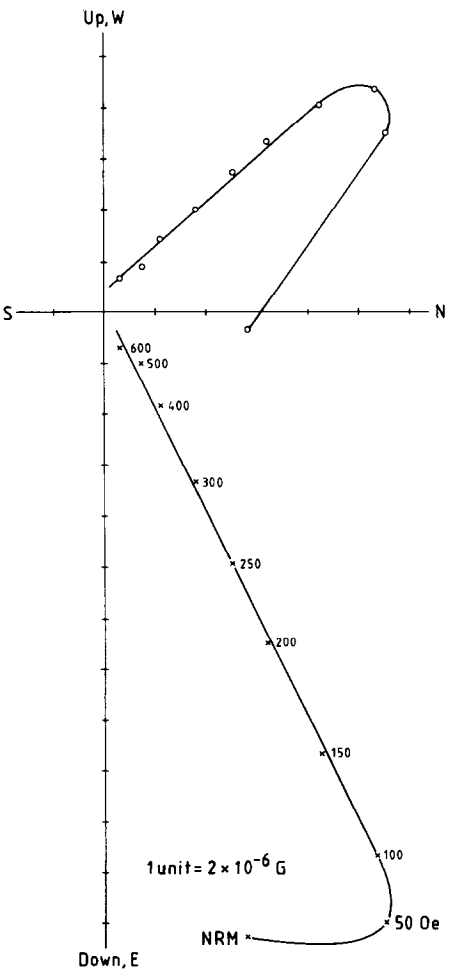


Fig. 13. Vector end-point diagram of NRM demagnetization of a typical amphibolitized diabase sample.

DISCUSSION

From the palaeomagnetic investigation of the various rocks and the sulphide ores, it appears that the magnetic pole positions (Table II) can be divided into several different groups. There is geological evidence suggesting that the different units are separated in time. All groups have significantly different magnetic directions (Fig. 14). Because the degree of alteration varies strongly from place to place, the various intensities of regional chloritization of basalts and uralitization of diabases do not seem to be the reason for the observed differences in magnetic directions. It is suggested that these alterations have been semicontemporaneous with the intrusions of the respective rocks. The diabase and the basalt and quartz diabase yield approximately the same latitude but differ about 70° in longitude. For reasons explained above, the dacite result is not included in the comparison. The position of the Permo-Jurassic poles from the Küre area differ notably from the European as well as the African data obtained from stable parts of these areas (Table II and Fig. 15). However, the pole positions indicated by most of the Permo-Triassic rocks and ores from the Küre area fall far from the zone covered by the Permo-Triassic poles of the African continent (Fig. 15). On the other hand, the poles of the younger rocks (Jurassic?) seem to be closer to the pole positions for Africa. A similar deviation

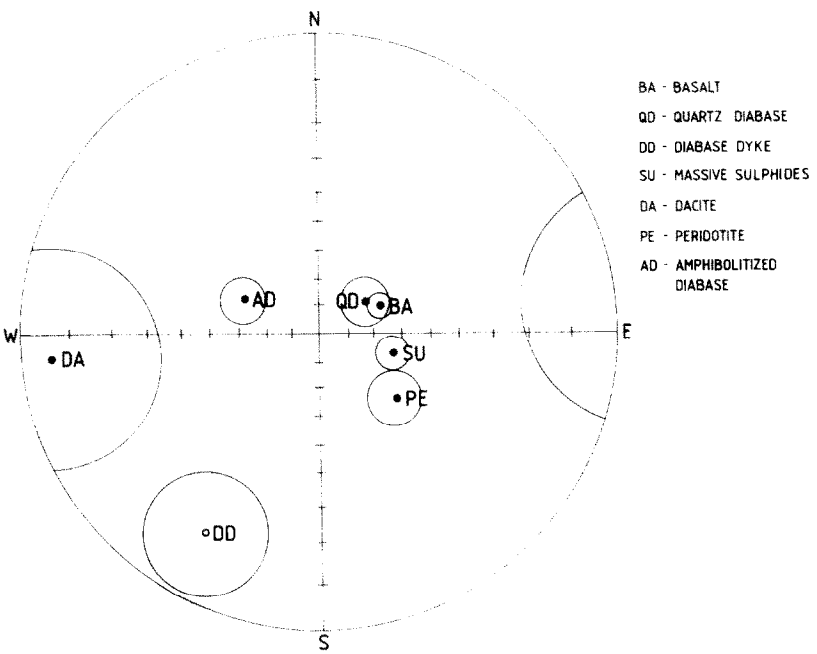


Fig. 14. Stereographic projection of mean directions after demagnetization. Dots = normal polarity, circles = reversed polarity. Large circles = angle of cone of 95% confidence circle.

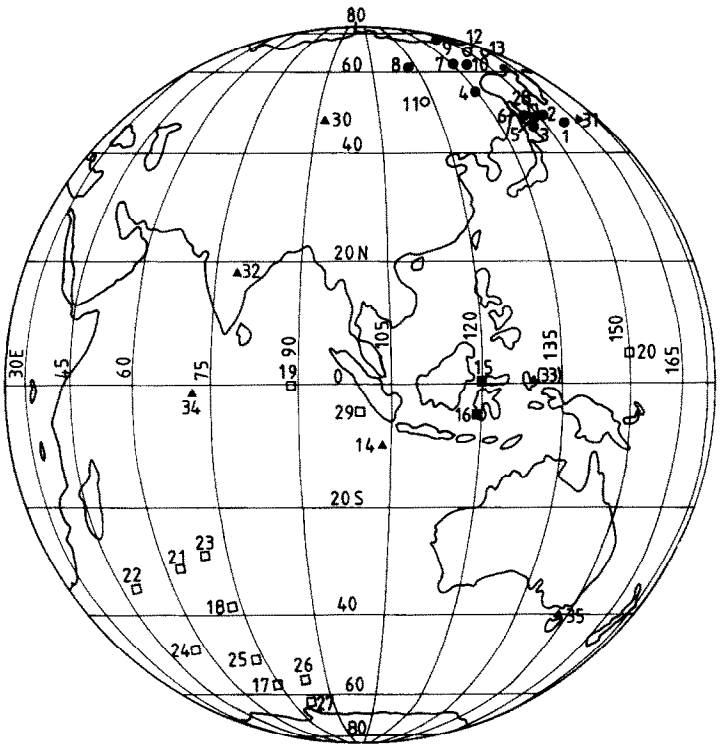


Fig. 15. Orthographic projection of Permo-Jurassic pole positions from Europe, Armenia, Minor Caucasus, Turkey, the Arabian Shield, Africa, and the Küre area. The numbers refer to Table II. Dots = Europe, circles = Armenia and Minor Caucasus, triangles = Turkey, filled-squares = the Arabian Shield, and open-squares = Africa.

from the African data is displayed by the Permian pole positions from the Amasra area (Gregor and Zijderveld, 1964), about 110 km west of Küre. The difference in latitude between the oldest rock (the basalt) and the estimated youngest one (the amphibolitized diabase) is about 85° (Table I). The pole positions of basalt and quartz diabase and diabase fit fairly well into the pattern of the Permo-Jurassic data of Europe and Armenia, whereas those of peridotite and amphibolitized diabase agree better with the Permo-Jurassic pole of the African continent (Table II and Fig. 15). This deviation can only be explained by the assumption that, since the Permian, the Küre area has been affected by large-scale rotational movements of more or less local character. For instance, a counterclockwise rotation since the Permian of about 70° with respect to the African directions can be suggested. The measurements indicate that, with the exception of the diabase, there are no marked deviations in inclination (Table I) between the different units. These range from 51° to 69° only, whereas the declinations deviate much more, i.e., from 51° to 293°. At the same time as the African and European continents (Laurasia and Gondwana-land) were

receding with respect to each other during the early and Middle Jurassic, the Pontic Ranges were experiencing intense subduction and later collision tectonics (Sengör and Yilmaz, 1981). According to Ketin (1962), Kimmeridgian movements of the Alpine orogeny characterize this part of the Pontic Ranges. Along the Black Sea coastal mountains, conspicuous folding and uplift have been observed between the Liassic shales and the Malm–Cretaceous limestones. The Küre area is situated only 70 km to the north of the large Anatolian transcurrent fault, which is still an active earthquake belt. The probability that this zone was an active dextral strike-slip fault at least since the Permian, in combination with the difference in magnetic latitude between the oldest and the youngest rocks, suggests that there was rapid post-Permian counterclockwise rotation in the considered region before the intrusion of the amphibolitized diabase. If this is so, the basalt magnetization direction requires a 70° clockwise rotation of the Küre area in the Upper Permian, in order to reconcile it with the African data (Fig. 15). It is hoped that further work on the Permian will provide more details about the structural history of the Pontic Ranges in connection with the North Anatolian faults.

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REFERENCES

Brinkmann, R., 1976. Geology of Turkey. Enke, Stuttgart, 158 pp.

Digico Computers, 1975. Balanced Fluxgate Rock Magnetometer Handbook.

Fisher, R.A., 1953. Dispersion on a sphere. Proc. R. Soc. London, Ser. A, 217.

Girdler, R.W., 1968. A paleomagnetic investigation of some Late Triassic and Early Jurassic volcanic rocks from the northern Pyrenees, France. Ann. Geophys., 24(2).

Gregor, C.B. and Nairn, A.E.M., 1971. Paleomagnetic result from Lebanon. EOS, Trans. Am. Geophys. Union, 52(4): 188 (abstr.).

Gregor, C.B. and Zijderveld, J.D.A., 1964. Paleomagnetism and the Alpine tectonics of Eurasia, 1. The magnetism of some Permian red sandstones from north-western Turkey. Tectonophysics, 1(4): 289–306.

Güner, M., in press (a). Geology and massive sulfide ores of the Küre area, the Pontides, Turkey.

Güner, M., in press (b). Geochemistry and tectonic significance of altered basaltoids from the Pontic Ranges, Turkey.

Güner, M., in press (c). A note on chrome spinels and ore minerals from the Küre area, northern Turkey.

Irving, E., Tanczyk, E. and Hastie, J., 1976. Catalogue of paleomagnetic directions and poles. Issue 4. Geomag. Ser. Nr. 6, Ottawa. Energy, Mines and Resources Canada Publ.

Ketin, I., 1959. Türkiye 'nin orojenik gelişmesi (The orogenic evolution of Turkey) Maden Tetkik Arama Dergisi, 53: 78–86.

Ketin, I., 1962. Türkiye jeoloji haritasi. Explanatory text of the geological map of Turkey. Sinop. Maden Tetkik Arama.

Ketin, I., 1966. Tectonic units of Anatolia (Asia Minor). Bull. Miner. Res. Expl. Inst. Turk., 66: 23–34.

Khramov, A.N., 1973. Paleomagnetic directions and paleomagnetic poles. Issue 2. All-Union Oil Geol. Res. Inst., Moscow.

Kovenko, V., 1944. Küredeki eski bakir yatagi ile yeni kesfedilen Asiköy yataginin ve Karadeniz orta ve dogu kesimleri sahil bölgesinin metallojenisi. Maden Tetkik Arama Mecm., 9 (2/32).

Martin, D.L., Nairn, A.E.M., Noltimier, H.C., Petty, M.H. and Schmitt, T.J., 1978. Paleozoic and Mesozoic paleomagnetic results from Morocco. Tectonophysics, 44: 91–114.

McElhinny, M.W., 1973. Palaeomagnetism and Plate Tectonics. University Press, London, 201 pp.

McElhinny, M.W., Briden, J.C., Jones, D.L. and Brock, A., 1968. Geological and geophysical implications of paleomagnetic results from Africa. Rev. Geophys., 6(2): 201–238.

Nairn, A.E.M., 1959. A paleomagnetic survey of the Karroo system. Overseas Geol. Miner. Resour. (G.B.), 7: 398–410.

Nairn, A.E.M., 1964. Paleomagnetic measurements on Karroo and post-Karroo rocks. Overseas Geol. Mineral Resour. (G.B.), 9: 302–320.

Opdyke, N.D., 1964. The paleomagnetism of the Permian redbeds of southwest Tanganyika. J. Geophys. Res., 69: 2477–2487.

Pavoni, N., 1961. Die nordanatolische Horizontalverschiebung. Geol. Rundsch., 51.

Phillips, J.D. and Forsyth, D., 1972. Plate tectonics, paleomagnetism and the opening of the Atlantic. Geol. Soc. Am. Bull., 83(6): 1579–1600.

Sarican, K., 1968. Etibank Karadeniz Bakirlari arama santiye sefligi. Bakibaba cevher yatagi arama ve degerlendirme calismalari raporu, Küre. Unpublished.

Sengör, C. and Yilmaz, Y., 1981. Tethyan evolution of Turkey: a plate tectonic approach. Tectonophysics, 75: 181–241.

Sirunyan, 1975. In: A.N. Khramov (Editor), Paleomagnetic Directions and Paleomagnetic Poles. Issue 3. All-Union Oil Geol. Res. Inst., Moscow.

Van den Ende, C., 1970. Secular variation in Permian Redbeds from the Dome de Barrot, France. In: S.K. Runcorn (Editor), Paleogeophysics. Academic Press, London, pp. 101–116.

Van der Voo, R. and French, R.B., 1974. Apparent polar wandering for the Atlantic-bordering continents: Late Carboniferous to Eocene. Earth-Sci. Rev., 10: 99–119.

Van Dongen, P.G., Van der Voo, R. and Raven, Th., 1967. Paleomagnetism and the Alpine tectonics of Eurasia, 3. Paleomagnetic research in the central Lebanon Mountains and the Tartous area. Tectonophysics, 4(1): 35–53.

Van Everdingen, R.O., 1960. Paleomagnetic analysis of Permian extrusives in the Oslo region, Norway. Skr. Nor. Vidensk.-Akad. Oslo, I: Mat.-Naturv. Kl., 1.

Zijderveld, J.D.A. and Van der Voo, R., 1973. Palaeomagnetism in the Mediterranean Area. In: D.H. Tarling and S.K. Runcorn (Editors), Implications of Continental Drift to the Earth Sciences, Vol. 1. Academic Press, London.